ActInf Textbook Group ~ Cohort 6 ~ Session 6 (Chapter 4, Part 1) ~ 3/18/20

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okay welcome back cohort 6 we're in our
first discussion of chapter 4 so
Andrew however you'd like to begin and
for everyone else any questions in the
chat or raise your hand or however and
let's jump into
four
perfect um yeah welcome everyone we're
going going over chapter 4 for the first
time with cohort 6 um for warning uh for
anyone new to active inference this
chapter is a bit more mathematically
involved than the preceding chapters um
but this is where we actually start to
relate a lot of the concepts and kind of
uh overview ideas that we've already
been introduced to we begin to kind of
map these uh more mathematically to
understand the computations involved
inactive inference
models um so more specifically this
chapter goes through the typical forms
of active inference uh generative models
will'll see both in discret and
continuous time um the relationship
between those models and uh the Dynamics
of minimizing their free
energy um at the very end we're also
briefly introduced to other things that
we'll see in later chapters including
inferential message passing so um I
would like to I I think this chapter
does well at highlighting the importance
of approximate inference um we've seen
before that the a lot of the underlying

assumptions in active inference uh
involves uh the minimization of free
energy now really the idea is that
agents would want to quote unquote want
to
uh minimize
surprise but because uh we need to
actually render the computation
tractable we instead are minimizing a
quantity known as free energy
um quick note that the mathematics
involved here draw from different
mathematical Fields primarily linear
algebra differentiation in in
multivariate calculus and the use of
tailor series expand
and some others um there's some nice
appendices in the back of the textbook
for anyone who's either new to those uh
areas of maths or need some kind of
refresher and would like to get a closer
look on their relationships specifically
with the computations involved here um
and so sort of the core here is that
we're introduced with a CA equation 4.1
we already know from previous chapters a
generative model in a in a certain way
can be reduced to a prior and a
likelihood um those fit right into basis
theorem that's equation 4.1 and so it
relates the prior and the likelihood on
the left hand side which you could just
replace with you know uh oftentimes
there's a notation the letter M to
denote a model and then on the right
side um those end up equaling a
posterior distribution which is your
your hidden States conditioned on your

observations
times um why or your your observations
so your your evidence and so you can
imagine whenever the agent basically
takes in new observations all these
quantities can be updated um and and
that's the kind of nice easier kind of
introductory view of how how this works
um as far as
rendering uh everything track
[Music]
um we also exploit something called
Jensen's
inequality um put in simple terms
Jensen's inequality says the log of an
average is always greater than or equal
to the average of a log um so we we can
exploit that and rewrite basis theorem
using
logs um and we use a different notation
rather than a capital P that might be
used to denote
an exact distribution we can use a q
which is used to denote an approximate
distribution um and and replace that in
there and so the whole idea is by
minimizing free energy as a kind of
proxy that is always greater than or
equal to surprise the agent again quote
unquote once to minimize free energy in
order to minimize its surprise if you
can get those two Quant ities to be
virtually zero and
virtually uh equal to one another that
that would be kind of the general
overall goal um and to kind of Zoom
through a bit more of the rest of what
happens in this chapter we get a further

breakdown of what comprises free energy
um it involves um using
Kack uh colak leer Divergence between
that approximate distribution the Q I
mentioned earlier and uh an exact
inference uh p and then subtracting uh
log model evidence from that um figure
4.3 excuse me 4.2 were introduced to uh
the kind of factor graph notation that's
often used in act building active
inference models um these are not yet
active inference models there're simpler
static perception models but they give a
nice kind of quick view of here are the
kind of things we'll be getting into we
have variables uh every single one of
these involves something we're already
familiar with which is a for a
generative model we have a prior we have
a likelihood so all of those models
involve the core components of a of a
generative model or agent right so um
from there we it it shifts into figure
4.3
um which will have very similar
components but we're now introduced to
active inference in that uh we now have
actions that the agent takes which then
impact uh State transitions right so
we've moved from figure 4.2 of just none
of the models in 4.2 have action
involved none of them have like
necessarily show some kind of temp
temporal sequence of things playing out
over time you know it's simply just
these more static perceptual models that
there's a nice example there where the
one in the top left can be modeled as

like um um a
disease and uh taking a diagnostic test
that then computes like the the result
of you know if if you uh administer a
test on a person who has a hit we're
trying to uh derive a hidden State here
marked as X do they have disease like
what is it or or do they have it or not
why is the result of the test right so
that's that's kind of a way of looking
at how these things are linked together
we have a prior which is the prevalence
of the disease we have a likelihood
which is um you know what is the result
of the test given the disease we can use
bases theorem
and what could be called Model inversion
to switch around see okay what is the is
there a disease given the particular
result of the test um strongly recommend
uh just kind of reading through those
couple of paragraphs to get a better
sense of what's going on there because
it's much more descriptively clear than
probably how I'm describing it here but
uh in any case yeah it's good resource
for getting started on these things um
and then yeah
4.3 the difference where we now have
active inference models is that we
actually have a temporal Dimension which
is we're now including State transitions
over time so you can read those graphs
in figure 4.3 from left to right you
know T minus one to T the current time
step to t + one which is the next time
step uh so the agent is basically taking
in those obser ations at the bottom of

that particular graph that that Daniel
has zoomed in on um you know we have
these linkages with the hidden States as
we saw before but now we have policies
impacting the state transition as the
agent uh engages in actions or sequences
of actions which is another way of
saying what a policy is in order to try
and uh the agent tries to impact the
changes in the states so that is kind of
the active part of active inference
right
um and then from there section 4.4 it's
a little bit more involved uh we can
take all of these ideas that we've been
introduced to
likelihoods um Transitions and so on we
can map those out to different uh
components of our models so now a
likelihood model um is now called an a
matrix or a tensor um a trans the
transitions model is now the B Matrix or
B
tensor the prior Over States are now a
vector called D um we also now have
preferences that the model has uh which
are uh in a c Vector super important for
um the agent actually realizing its
preferences or expected observations
which is an idea we're already
introduced to in uh previous chapters so
the idea here is that we're able to just
kind of map a lot of the more uh
verbally described ideas that we've
already seen in preceding chapters to
these different components of a model
and then seeing how computationally we
can and uh minimize free energy over

time couple other ideas that are brought
up but I won't go into too strongly
right now because I want to allow time
for questions
uh in in discussion are expected free
energy or otherwise known as G um that's
related to planning this is where uh the
the agent ju doesn't just look at past
observations or current observations to
minimize its free energy but it also
wants to realize its preferences and so
it has to kind of plan what it will do
in order to realize those preferences or
goals that it already has
right so G plays in in that that the
agent actually wants to kind of plan
ahead of time what what should it do um
so the C Vector of preferences plays
into that um you can look at the way
that the agent chooses different actions
or policies to take as um it's actually
Computing expected free energy for each
policy and then it scores the policy see
should I do X should I do y or let me
revise that because that the the
notation I don't want to get mixed up
should I do policy number one policy
number two policy number three well
let's score the expected free energy for
each of those policies and whichever one
has the lowest expected free energy at
the end you go with that policy and you
go with the actions that that policy
entails um and order to impact the
environment in order to impact the state
transitions based on your transitions
model which basically is how do you
think your actions are going to impact

the environment around you
um even more quickly some other Topline
topics that were introduced to Markov
blankets we've already seen some
references to those uh in previous
chapters um there's a nice box uh that
descri rpes more on Mark of blankets and
not just how Mark of blankets are a kind
of
conceptual concept that separates an
agent from its environment but
furthermore kind of the maths uh
underlying what a Markov blanket is
about statistical independencies and
dependencies between variables and so on
we get more on continuous time models
which are frankly much more
mathematically involved I would say um
involves more uh in the way of
differentiation in looking at temporal
derivatives and applying tailor series
expansions uh a concept called
generalized coordinates of motion that's
involved as well as the Plass
approximation and uh finally at the very
end of the chapter we're briefly
introduced to what's called basy and
message passing just kind of how um if
we think about these models as graphs as
we've already seen um messages are being
passed around between nodes via edges um
and that has uh significant amount of
neurobiological plausibility behind it
theoretically and and and through
continued empirical study and so that's
kind of the that's the kind of
neuroscientific or
neurobiological uh backing to to being

able to say how a lot of this works and
for anyone's especially interested in
message passing we we'll see a lot more
of that in chapter
five um I would also recommend to anyone
more interested in uh planning in these
discrete time models with the A and B
and C and D components and so on if you
want to skip ahead to chapter 7 um there
that chapter is entirely centered on
discrete time models um it'll have more
breakdowns of these different terms that
are involved in Computing G like we see
terms like like expected ambiguity and
risk in this chapter there are other
ways that you can break down how we
compute G and it's it's nice to have
these kind of words like risk or
expected ambiguity that maybe gives a
better intuitive sense of the maths
involved and how it's useful how it
plays into the explore exploit tradeoff
uh Model Behavior anyway I I I feel like
I've already covered the gist of of
what's going on this chapter and maybe
even overs spoken so uh if anyone has
any questions um thoughts on any of this
or any other particular topic they want
to get into um please feel free to uh to
raise your hand or just unmute yourself
and go for
it yeah I have a couple questions
related to the statistical side how
would this relate to calculating
something like the r not like we look at
in epidemiology where you're looking at
a predictive state with an N value of
zero I'll give a short answer there

there's probably multiple ways you could
do it but I mean one way to handle it
would be treat the true AR not as a
hidden State and then you make
observations so you have an
epidemiological infant trying to
estimate an underly Ling parameter that
could be the temperature in the room
with the thermometer it could be the rot
with the sampling from the noisy
tests that's kind of the general
approach it's it's what SPM does you
have the underlying neural activity
unobserved and then you have
neuroimaging data which is being caused
by very um nonlinear relationship with
neural activity like with the blood flow
that's a general process of perceptual
inference is pretty much separate where
you're really the underlying variable
hidden state that you're trying to
estimate and then separate that out from
the observations you're getting and then
you load up onto the a how the hidden
state is related to the observation
synchronously then you load up onto the
B how that hidden State changes through
time and then you can layer in the
policy okay can you do that recursively
with elliptic curve then so would you
step back over so you would have to do
linear step function back over through
right again not sure how the curve you
mean
like the so I'm just thinking in terms
of the math with like the heavy side
step function so if you're going down
from a three state to a two

State see how you have like you're
almost jumping that zero state but
really what you're doing is sinusoidally
passing through it with an elliptic
curve in a way but you're passing
through a negative with a positive
direction sketch out an agent that's
doing that or having it done to them and
then try to make it look like figure 4.3
and then you'll probably be close to an
answer
okay
thanks for the summary Andrew this is
one of this is the dense
chapter
um
yeah spending time
with yeah chapter one introduction two
and three low and high road getting two
active inference chapter four it's
active inference chapter five
neurobiology case studies of active
inference that's the first half of the
book and then chapter six recipe for
making it yourself chapter 7 and 8
revisit sections 4.4 to
4.5 nine bringing in data chapter 10
summary
so we get a significant amount of kind
of back and forth between kind of more
verbal descriptions of the the ideas and
Concepts but we have to come back to the
maths right that's That's essential and
that's I think that's what chapter 4
does is just quickly like give you a
broad stroke of here are the maths
underlying a lot of these ideas and how
the Dynamics play out

um Jr made a really nice comment in the
Discord a few days ago this is the a new
paper active data selection information
seeking and here there's a slightly new
and
different uh
notation Lambda operator introduced to
indicate either summation or integration
depending on whether we're dealing with
continuous or discreet
variables so here we're approaching that
in the time setting but it's kind of
like this comes up a lot like whether
the variable is continuous or discrete
and then whether time is being treated
continuously or discreetly those are
Options under the umbrella of active
inference
models so the actual kind of pseudo code
is not predicated on the variables being
continuous or discret M
that to me intuitively just
seems really nice and allows for a more
direct
relationship between the continuous and
discreet variables because yeah this
chapter 4 here as well as comparing
chapter 7 and chapter eight it's like
this book The Textbook kind of presents
discreet and continuous I mean of course
free energy minimization is an in lot of
other concepts are shared between the
two but I mean we get a kind of sharp
distinction between using discret and
continuous variables all the way down to
the very notation being different your
S's and O's for discret versus x's and
Y's for for continuous and so on so it's

nice to see an equation that can kind of
relate or or compound the two to where
you just need to kind of know if it's
discret continuous and then similar
process for each while knowing of course
that discrete variables you're working
with Point masses and summations as
opposed to Integrations and so on with
continues that's another point to be
taken for the ontology
sure
um Peter or or anyone sure thank you
Daniel um so I loved this chapter I
thought it was really cool um I'm uh
pretty excited by the uh the Lambda
notation that was just described I was I
was unaware of that and uh there's a lot
to think about there um in some ways
this was the chapter that uh I was
personally like least comfortable with
in terms of my my ability to sort of
grock everything that I need to get um
my personal background is mostly on the
neurobo side and I have much less in the
way of mathematics um I was wondering so
I haven't checked out the um the
references at the back yet so maybe
maybe that's actually going to answer my
question but I was wondering does the
active infant group have any kind of
like a learning path for the supporting
mathematics uh that you would need to
really be able to be really um kind of
fluent with this chapter and like for
example be able to plug in values and
and do things with the model like do we
have any kind of a guide of like hey if
you want to learn you know um the Taylor

series expansion here's where to get it
if you want to learn linear algebra like
here's a particularly good text um
because I'd be interested to sort of
take that stuff on over
time thanks yeah um in the math learning
kind of
subsection this is you know quiescent
till um or when
people don't want to do it but then as
people want to do it like anyone's
welcome to just say this time UTC I'm
going to do this uh like in Discord or
however or just make the edits directly
here this math learning and
resources has it at least
many
um I think the language models are
really good at this right now this kind
of like um like undergraduate level math
which is hard that is a good specialty
of them and just put and then another
possibility for with the learning is
from the um
equations here all the equations have
been put into
latch then this can just be so even if
you're working with a language model
that doesn't have the ability to put in
a screenshot you can just put in this or
look at what we have here and then look
at the explanation why would a high
schooler be interested what's the pseudo
code where is it important what are the
variables these are just tenative but
they do um even where they contain like
some like I guess factual you know
omissions or or

errors they they give it the
style of what information is in the
equation and just like 1 plus 1 equals 2
1 plus one equals question mark like you
could spend the rest of your life truly
pondering it so there's never going to
be a moment okay I'm done with B
equation I finally you know don't need
to wonder about it anymore but then
knowing that it will never be fully
resolved getting to kind of just like a
stopping point with feeling like you
don't know at all what it
does and then just seeing a bunch of
related natural language um descriptions
and then also our
handwritten natural language
descriptions which anyone can ALS so add
to sometimes just seeing it written out
linguistically instead of only
symbolically can help especially because
then you can replace x with hidden State
and Y with
observation and then from there that can
translate to other languages but also
you could replace okay hidden state with
temperature observation with
thermometer and then this starts to look
more like the probability of the
temperature times the the thermometer
given the temperature
and so then it's like easier to jump
between a tangible example um and all
these other like kind of aspects of the
holistic math understanding and like
Jonathan shock and Ali blue others have
um like also thought about how
to what what kind of math is important

to lead up to the textbook group and in
the in the appendices they do mention
what they see as the important
background
areas I think it's
a linear algebra so doing operations on
matrices um adding and multiplying
products on matrices tailor series
approximations this is important for the
continuous time this is where you take a
point and then you try to extrapolate a
continuation around that point
variational calculus and stochastic
Dynamics
like this
appendix may still not be the perfect
resource to learn those four topics but
the fact that they listed them out as
the four topics than going to any like
highly rated online course
or like popular videos on these topics
is probably pretty helpful linear
algebra very much is and stochastic
Dynamics or just statistics
I think also is critically
important it's fantastic thank you it's
uh Beyond helpful really appreciate
it Peter I'm in the process of learning
all of this myself so I'd be willing to
work with you if you wanted to do um
something as well um within Discord or
something through
Cota sure that'd be great
thanks are you uh sorry go good are you
on the Discord just so I can uh yeah I
just joined a couple days ago okay
great there is a textbook group
Channel um you can you can try to join

in or just post somewhere if you can't
find it or anything but then we can just
kind
of keep that discussion alive and
um like yeah
knowing knowing what this
symbolizes
helps there probably is a point of
diminishing return with learning any of
the background maths but also they're
their own reward so they're fun to
learn I might throw this into the um
Kota as well but just sharing if anyone
is familiar with um like corsera courses
for you know for me I I've had to do a
lot of learning async SL from The
Institute and I I really like this one
um just because um you know it has a
couple courses there it's very intuitive
and uh especially for anyone who wants
to move in the direction of like
implementing and learning with um python
it mainly focuses on that but yeah the
first course is linear algebra but in
the context of like we're learning
matrices we're learning how to multiply
them and so on um and then by time we
get to the second course it's multi very
at calculus and so all of a sudden
you're learning about like gradient
descent uh with multiple variables
involved so just the first two courses
of of that link I shared um I find
immensely helpful for at least as a
starting point it just walks you through
everything
so
yeah depending on where someone's at

with their programming or engineering or
anything it can help just to think of
these as variables like s can be kind of
H treated with one handle as like a
mathematical variable and if you want to
go into the um like the abstraction and
the analysis and the category Theory
then it is going to be helpful to just
pick up S of t with one handle but then
at the same time when you actually write
the pmdp code s is going to be like a
matrix of a defined shape so then going
to the pmdp documentation and examples
and just seeing like okay what shape is
AB c d okay that's the whole agent
that's the active inference agent that
can help ground it that's not the
general case but seeing how um
minimal a small case can be is helpful
to know like what a complete um
generative model is and kind of
calibrate how challenging different
kinds of phenomena might be to model
yeah pmdp open source package discreet
time but this is a good um package
that's really focused to like getting as
fast as possible to the discreet time
agents whereas RX infer as we're
exploring week after week um is en and
continuous and all these other things
PDP gets you really fast to the PDP like
chapter 7 model um there are
great demos including that you can just
open it up in a
notebook you can copy the notebook or
use this version and just like run it
right here and see the full active
inference model you can also clone it

locally
into like cursor.
sh with a code enabled development
environment and then you can chat with
the code base and ask questions about
like pmdp or any other
package so this can be extremely
helpful
yeah I think that's great um I would say
for for people who are just getting into
this and thinking about how to set up a
model as well some things uh if you want
to start with pmdp or
otherwise um and again if if you want to
jump to modeling I would strongly
recommend having a and you have the time
for it have a look at Chapters seven and
eight that'll definitely go over more
like more specifically how to construct
models um for anyone who's thinking like
okay I have a particular project how do
I map the variables to these different
components of a a pomdp for example you
could have a look at chapter 7 it it'll
get into things like okay uh we have
States we have observations like two
crucial parts of of a model um well
maybe I have different forms of data
that I'm bringing in like for example
someone doing something um more
physiological and clinically uh based in
a lab or something maybe you're
collecting neuroimaging data in addition
to heart monitor data so suddenly you
have two different data streams coming
in um you can map those as different
like observation modalities is what
they'd be called and basically you'll

just be constructing your your matrices
the A and the B and so on that account
for like these different data streams
that account for different hidden States
that you're trying to inferring uh
trying to infer so on um pmdp is great
for that too uh if you are wanting to
use python or at least using it as a
learning tool um because all of these
examples there's the T one that Daniel
scrolling through there's another one
that sets up a two armed um Bandit model
which is like uh an agent who um you
know picks from one of two like kind of
slot conceptually what are like slot
machines and um so you have it it shows
you how to like oh you have multiple
forms of data or observations that are
coming in you have multiple hidden
states that the agent is trying to infer
you will set those up as like matrices
or tensors um and that's kind of where
the the linear algebra plays in and
being familiar with Matrix um operations
becomes really important but uh it's
it's really nice yeah David you have
your hand up uh yeah in regards to the
choice function there would that apply
to what we covered earlier in terms of
the Lambda function or the Lambda
operator rather
see um okay we can we can look back at
the lamba but about like the behavioral
selection itself let's find like kind of
exactly where it comes into
play
um okay okay so um in this this is the
uh tutorial

one there's going to be
uh they they first have a simpler
version with no
planning so here's an active inference
loop with no
planning then planning comes into play
so big big question um like AR ing when
doing the modeling is like are we
working with something that
plants um and even behavior that kind of
like seems plan likee might be
underpinned by simply heris sixs like a
flock of birds you might not need to
appeal to it doing a long-term planning
for its coordinated movement you might
just be able to explain that with single
time step behavior of the birds so just
because something is like acting
adaptively through multiple time scales
doesn't mean that it needs to be planned
that's a um but you may still model
something from the outside as if it's
doing
planning um whether or not it is map and
territory and all that okay now given
that here's here's where when you're in
um the planning setting here's where the
the choice comes into play
calculate expected free energy of
actions so here we we actually are using
expected free energy G and it it is
calculating the expected free energy
based upon essentially all the variables
that come into
play and then they um so it takes in the
policy prior which is like e or habit
which is just your your before thinking
about

what is your habit so if you're doing
like a fair teas rat it might just be
50/50 or if it's a um if it's a teamas
rat but it has like an 8020 bias for One
Direction um then each of the
policies get um
reweighted the policy the policy
variable sums up to one and then here
they're reweighted to update to a policy
posterior and then here chosen action is
sampled from the posterior
earlier Andrew also said you could just
pick the lowest one or you can sample
from this distribution and there's other
things you can do so expected free
energy isn't a choice function itself
it's a reweighting of the likelihood of
policies that then enables you to take
different kinds of choice functions
secondarily so would policy Q be in
essence the quiescent point of
Pi which one so the the so where you
have
qore Pi is the soft Max at negative
G so would that be like a horizon Point
essentially where you're measuring
at um not or measuring to a distance at
soft Max the the time Horizon is Big T
that so here this is a five time step
simulation and here policy length is
considering all um combin ations of
policies of two affordances so if you
can go up down left right four
affordances then there's 16 policies of
length too and you have a range of four
values with the N actions as the okay
yeah the actions at each moment are the
affordances policies are sequences of

affordances for a given
time
um and then playing around with these
variables is sweeping across them making
like summary diagnostic statistics about
portfolios of
simulations is when you start to see
interesting things okay great yeah I've
been kind of modeling similar to this
and I've been seeing similar things so
I'll start working with this a bit more
to solidify things perfect thank
you
cool and yeah as uh as Daniel mentioned
earlier for anyone who's more directly
interested in in modeling um again I'm
I'm personally trying to learn both PDP
as well as our RX infer at the same time
uh two different programming languages
because RX infer is based in Julia but
uh we do have an ongoing Institute
project Thursday mornings um it's the RX
infer project and with great is that um
one of the folks from bias Labs who's
developing the RX infer package um is
actually directly involved with that
group so we we have some some nice uh
communication with one of the developers
basically um and there we are um going
through different examples right now
that ARX infer like those folks have
directly made available just like we
were going through an example for PDP so
they've done the same for ARX and fur um
there's a nice kind of reci reciprocal
relationship going on because we have
the opportunity to tell the
representative from bias Labs like hey

we think you should you know maybe
update your documentation to make it
clear on this particular point or
something like that and then they help
us with with the learning but yeah
anyone who wants to kind of get more
involv with that um not to mention the
the ARX and fur package seems to offer
some capability ilities Beyond uh maybe
what other methods allow for so um we
were able to look more at like
continuous time models which are
certainly uh in certain ways more
mathematically involved with all the
Integrations and so on so yeah this is a
nice um example here like this is trying
to model um a car that needs to have the
it needs to take the right actions to
exploit um like its velocity and
acceleration in order to move up a hill
landscape to reach a goal that it
otherwise couldn't if it just put the
the pedal to the floor so to speak um
and so the agent like learns what um
what it should do in order to reach that
goal um but yeah so they're just like
toy examples but um yeah super cool yeah
stuff and I think useful for yeah if you
have your own project that you under
want to undertake using this as kind of
like a foundation for that Daniel you're
going to say yeah yeah yeah this this is
a different syntax different language
Julia um than python but with with many
similarities and again like even with
limited code familiarity it helps
calibrate like where someone is at like
this is just a coin toss this isn't even

like it's not a coin tossing Quantum
agent this is just like literally a coin
being tossed but this can help calibrate
learning in terms of
like what does it take to make a coin
toss
and then okay here's a slightly more
involved example with a hidden Markov
model but um you know then this this is
um and then we're also going to bring
this back into the um textbook Group by
looking to
replicate PDP and textbook examples in
RX and
fur because like although it can feel
and and it is true on one hand that it's
like more more more flavors or like more
material or more variants it's actually
a very important decision that they took
in the textbook to have like figure
4.3 like they're not in they don't
introduce just the discreet time and
then later go to continuous time they
introduced them jointly to emphasize
that there isn't only one way to treat
time in active inference so then that's
how I've been approaching it too so
that's awesome cool yeah I mean it's
like there's I I don't know if there's
any um any oneway streets like if
there's one there's another one and so
having there be multiple
languages it's like not about one person
being able to generate all the scripts
for all the languages for all the cases
you know in a moment but basically code
models can already do that first off and
then so we're just able to look at that

and then try to um connect the dots and
and think what's this deeper space
that's able to have multiple Julia and
multiple Python scripts corresponding to
a
teas and then how do we make sure that
that kind of like broader latent or dark
space that we're thinking about is
active
inference and what makes something
actually an active inference model as
opposed to just like any other kind of
broader input output cybernetic or
control or reinforcement model which
might be totally valid and adequate for
a setting so there's nothing wrong with
like not making an active inference
model it's just that there is a
difference between it being an active
inference model versus being like a
reward learning
model and right so I kind of see that
sorry no go ahead so I kind of see that
also as like being a parameter of the
time frame or the frame that that
constraint is set in and so if you're
using the language in that sense as a
filter it would be applicable to that
model for example in that discrete time
range where it may not fit to another
model with the same kind of parameters
due to you know other other separate
constraints for
example you mean like kind of using the
um validity of the active inference
ontology
to kind of check whether the script is
an octave inference model yeah but in a

way by being able to apply it as a
filtering model without energy it it
kind of reduces the friction on the
surface doesn't
it so if the if the language that you're
using within that frame works with the
frame um without it adding too much
context because of you know the
parameterization versus dragging
dropping that filter into another model
where the Lang which doesn't fit um you
would have to use a different model but
that doesn't make the first model not
valid would that be a correct statement
in that
context I I think so I mean I I don't
know this too broad but making model two
cannot make model one invalid model two
can yes and it might be be yeah it might
be fewer parameters or more or more or
less accurate but but it can't
invalidate it's just merely another
addition like into the blockchain of
everything and then in terms of the
ontology and the
variables I mean you can just look at
the
variables
oh added more comments here but um if
you've defined if you've looked at
figure
4.3 and this is the rough setup of your
set you then you can ask I is there
something that is here that I don't
don't
have and then is there something else
that I have and that and either of those
if you make it simpler than

this um you might be dealing with a
special reduced
case if you add more things I mean that
is pretty much all of like the research
advances in active inference is like
well what if we added a variance um you
know parameter here or what if we added
a nesting here so it's like more can be
add added um and and hence the textbook
just like try to throw the fast ball
down the middle and show it in its like
plain way and then to be able to map
anything to the plane
version if you can bring it back and if
it's a valid return to the to the kernel
you know then it's like oh it's a
heavily modified Linux kernel but not
everything is a heavily modified Linux
kernel something else might just be a
totally different construction so it
might be compliment it might be better
might work well with in a computer
system but it it wouldn't make sense to
say that it was a modified Linux kernel
unless you had started with either
pruning down or adding to the Linux
kernel okay perfect yeah you don't want
to end up in Temple OS when you're
looking at
Linux I would just also briefly add to
that that point if like just for
inspiration or otherwise but if you
wanted to kind of just literally just
flip through not even necessarily read
but for the time being flip through
chapters seven and eight and just look
at the other figures going on in there
they give um many more ideas of like how

you could construct this from the standpoint of like looking directly at these this kind of like graphical notation um we've been talking the RX and for how it seems great to be able to use graphical notation in the first place one just for being able to track all the different variables and how you're setting up the model but then also two just like steeming on like what what do you need how does the model need to be changed or or otherwise um like in seven chapters seven and eight we get more into discret and continuous time models we also are introduced to hierarchical models you can add a second level um probably one of the most involved graphs is yeah these um like Daniel has pulled up I would say especially figure
8.6 um is is pretty great what it does is that you have a hierarchical model at the top level you have the PDP like discreet uh time uh model setup and then the the bottom half of that graph is actually um using continuous variables so so it's not just hierarchical it's also like a hybrid model so you're in including both types of variables in there uh you could almost think of this as like at the lower level that bottom half you could have something like um uh continuous time neuroimaging data and then at the higher level there's some kind of discret uh hidden state that you're using that that real time observational data to then uh infer like you know so it's um yeah it's just

steaming modeling doing the mapping from
uh you know reality to to variables on a
graph um awesome yeah this all uh all
fits together so yeah I might have to uh
join you guys on Thursdays that'll be
interesting thank you yeah or or um or
the the RX in foras in the chat also on
the math background so the SPM
statistical parametric mapping this is
what Carl friston at all were working on
in the neuro image group at UCL in the
preac and FP days and continuing um it's
a Matlab toolkit it continues to be
developed and the documentation is all
online that being said this 2007
textbook is very informative it has like
40 or 50 chapters but they're Standalone
short
chapters um that that don't really
reference each other it doesn't Focus so
much on the ACT and control theory side
because they're focused on like the
neuroimaging setting of taking in um
multi- sensor data doing the Sensor
Fusion doing the dynamic caal
modeling I find it really informative
because it it quickly reviews and and
uses parametric statistics
non-parametric statistics and basian
statistics so it's very very strong for
learning about this setting which is
where you have a prior on a hidden
variable and then you're getting
observations and then the hidden
variables changing through time like
neural
activity control theory then then
injects into selects out a slice of B to

control how the system evolves and so
that's why that was brought into play
because if you're only observing the
neural system then you only need this
but then if you're going to be selecting
behavioral um stimuli or you're going to
be doing like transcranial stimulation
or something like that you need to have
um a bir directional interface but in
terms of like the basic
statistics and weaving together
parametric non-parametric basian this is
like one of the best textbooks okay
awesome I think I've read some
selections from it but I definitely need
to uh I might pick up a copy of it I
don't know how much the physical version
costs but um I'm sure that there's a PDF
of this one we we probably have it and
also the online documentation is is more
updated okay and there's and there's
also tutorials for
SPM um so even if you don't do
neuroimaging I've never I've never been
in an MRI um slash analyze the data but
still just reading through this because
these machines are creating a lot of
data and it's a very transferable
setting okay great yeah I'm interested
in the noise parameters on that
too yeah and comparisons the kinds of um
the kinds of Statistics that you need to
account for like within and among
participants in a clinical study and
statistical power calculations like
that's really where it comes down to it
like trying to understand the
statistical

formalisms in their nals 1 setting is
almost like I don't know if ironic is
the right word but the whole point is
we're using distributions so that we can
take in multiple data points so it's not
meant to just be like kind of looked at
on the
Shelf it's meant to be like played
around with different priors and testing
how different observation sequences lead
to different updates so that's kind of
how I've been approaching it as well is
testing as n equals z is equivalent to
one which is why I asked the question in
regards to the Lambda is a choice
function of Z or one um but that would
so I've kind of been approaching in
terms of like at t0 n zero at T1 And1
and you have expansion from that set
back to and will always still be the N1
Plus or n minus 2 and that sends back to
just what we had started talking about
earlier but
yeah there's a lot there that's kind of
like when when the when the when the
mouse is
planning it's it's now it's t equals z
but also it's the next time step but
also it's an arbitrary time step but to
get lost in the fact that it could be an
arbitrary time
step it also has to be through other
frames
right well um next time later in the day
let's just begin with whatever questions
people have or other added questions to
the table and upvote the questions that
you like and we'll start from the top

next

week

okay thank you all awesome thanks

everybody thank you byee thanks folks

take

care